BADR ALBDULMAJEED

Graduate Protection and Control Engineer

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering graduate seeking a graduate Protection and Control Engineer position. My degree work centred on power system protection, including an IDMT overcurrent coordination study for a 230 kV generation and transmission system, phase and earth-fault protection modelling for an 11 kV network, and load-flow studies in PowerWorld and MATLAB. An 18-week industrial internship added Siemens PLC training and a relay-isolated 24 V DC sensor interface. I work methodically and check results against IEC and IEEE standards. UAE Golden Visa holder, available immediately.

TECHNICAL SKILLS

- **Protection:** phase and earth-fault overcurrent protection (ANSI 50/51, 50N/51N); IDMT relay setting and coordination; CT ratio selection; time grading; time–current curves; IEC 60255-151 and IEEE C37.112 characteristics; fault analysis.
- **Power systems:** load-flow analysis (Newton-Raphson, Gauss-Seidel, Fast-Decoupled); transformer and alternator testing; transmission line performance; high-voltage testing; microgrids with solar PV and battery storage.
- **Control and automation:** Siemens S7-1200, S7-300 and S7-200 PLCs; TIA Portal and SCL; relay logic; relay-based signal isolation; 24 V DC control circuits; LabVIEW.
- **Software:** MATLAB, Simulink, MATLAB App Designer, PowerWorld Simulator, LTspice, DIALux evo and SolidWorks; programming in Python and C++.

ACADEMIC PROJECTS

Overcurrent Relay Coordination Study, 7-Bus Generation and Transmission System – *MATLAB App Designer*
Determined the plug setting, time multiplier setting and operating time of ten IEC Normal Inverse relays protecting three generators (13.8–18 kV), their step-up transformers and the 230 kV lines. Selected each CT ratio, confirmed 0.3–0.8 s time grading between every primary and backup pair, and built a MATLAB application that plots all time–current curves (maximum calculation error 1.00%).

Phase and Earth-Fault Protection of an 11 kV Distribution Network – *MATLAB/Simulink*
Modelled a 33 kV grid supplying a three-bus 11 kV radial network through a delta–wye transformer, protected by phase and earth-fault overcurrent relays. Simulated line-to-ground, double-line-to-ground and three-phase faults, compared six IEC and IEEE inverse-time curves (0.21–0.47 s), and confirmed backup operation with a 0.2 s margin when the primary relay failed.

Power Flow Analysis of a 10-Bus Network – *PowerWorld Simulator and MATLAB*
Solved the network by the Newton-Raphson, Gauss-Seidel and Fast-Decoupled methods, with Newton-Raphson converging in three iterations. Adding 35 MW at Buses 3 and 6 lowered Bus 3 to 0.969 pu and raised total losses from 6.15 MW to 9.80 MW.

Additional laboratory and project work

- *Electrical machines:* ratio, open-circuit, short-circuit and load tests on a transformer and a three-phase alternator.
- *High voltage engineering:* impulse voltage generator waveform shaping and a three-stage Cockcroft–Walton multiplier, with results checked against theory.
- *Final year project:* self-supervised Vision Transformer for PCB defect detection (F1 score of 0.83, 22 classes).

PROFESSIONAL EXPERIENCE

R&D Project Engineer (Internship), *Control Easy Technology Sdn Bhd, Selangor, Malaysia* **March 2025 – July 2025**

- Designed a galvanically isolated interface using an OMRON G2R-1-S relay so that a 24 V DC industrial proximity sensor could safely trigger a 3.3 V Raspberry Pi input, then wired and tested both circuits.
- Completed hands-on Siemens PLC training on three systems: an S7-300 in a SCADA simulation, two S7-200 PLCs switching smart relays for pump monitoring and control, and an S7-1200 driving a 3-axis pneumatic pick-and-place arm, all programmed in SCL through TIA Portal.
- Delivered five machine-vision quality-control applications, each with a PC version and a live Raspberry Pi 5 version, replacing a proprietary C# tool with open-source software.
- Wrote the company's 259-page Open-Source AI Deep Learning technical manual, covering dataset preparation, model training, GUI development and Raspberry Pi 5 deployment. Rated "Excellent" (Grade A) by the company supervisor.

EDUCATION

Bachelor of Engineering (Honours) in Electrical and Electronic Engineering **2026**
Asia Pacific University of Technology & Innovation (APU), Kuala Lumpur, Malaysia
Relevant modules: Switchgear and Protection; Power System Analysis; Generation, Transmission and Distribution; High Voltage Engineering; Electrical Machines; Power Electronics; Control Engineering.

CERTIFICATIONS AND LANGUAGES

MathWorks: MATLAB, Simulink and Machine Learning Onramps; Core MATLAB Skills learning path (2026).
Languages: Arabic (native), English (fluent).